

Document field detection with computer vision

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Abstract—In this paper, we present a computer vision approach for the processing, enhancement, detection and text recognition of document images. The main goal is to show that we can use computer vision techniques on photographed documents to make field values detection automating the process of information registration in various situations showing a good amount of accuracy in the fields values recognition.

Keywords—Computer vision, image treatment, digitalized documents, text recognition, field segmentation.

I. INTRODUCTION

With the growth of digitized documents, it is essential to ensure that the captured images maintain the necessary readability for future processing. However, issues such as low-quality camera images, poor lighting, and the presence of noise can compromise the clarity of the information in documents. In this context, computer vision emerges as a powerful tool for image processing and enhancement. Through preprocessing techniques such as the use of filters and edge detection algorithms, it is possible to improve the visual quality of documents, making them easier to read and store.

With the use of computer vision also makes possible the automatization of information registration using just a camera and taking a picture of the document. This paper shows how that is possible by preprocessing the image and using EasyOCR[1] python library to extract the information of the document making the whole process of registration of these information easier and faster because we able to take pictures of a bunch of documents and process the information without interference of a human.

II. SYSTEM DESCRIPTION

The developed system aims to analyze and process images using computer vision techniques. The executed steps follow a structured flow, starting with the import of necessary libraries, such as OpenCV for image processing, NumPy for matrix operations, EasyOCR for detection and text recognition and Matplotlib[2] for visualization of results.

Next, image reading is performed, which is initially loaded in color and subsequently converted to grayscale, facilitating processing in the following steps. In the pre-processing stage, a mean and median filters are applied to smooth the image and reduce noise, promoting a cleaner base for subsequent analyses.

In the edge detection phase, different algorithms are tested, with Roberts appearing to be the most appropriate due to its ability to identify contours with good precision and low computational cost. This edge detector is capable of highlighting abrupt intensity transitions in the image, which usually correspond to boundaries between different regions of interest. Soon

after, segmentation takes place, where the image is binarized through thresholding and then morphological operations, such as closing, are applied to correct flaws and join nearby regions. This flow allows for precise visual and quantitative analysis of the elements present in the image, demonstrating the efficiency of the techniques used in the system.

The first step is to convert the image to grayscale, which makes it easier to apply filters and extract the text.



Fig. 1
ORIGINAL IMAGE



Fig. 2

IMAGES IN GRAYSCALE, BLURED USING 3X3 KERNEL AND MEDIAN FILTER USING KERNEL 3X3

After applying the mean and median filters, which help smooth the image for better text recognition, we can now apply a threshold filter to convert the image into a binary format.

This helps in recognizing the letters in the text. Threshold values ranging from 120 to 200 were tested, where pixel values below the threshold become black, and those above become white. Among the values tested, a threshold of 200 provided the best result, offering a good balance between accurate letter recognition and the removal of irrelevant information.

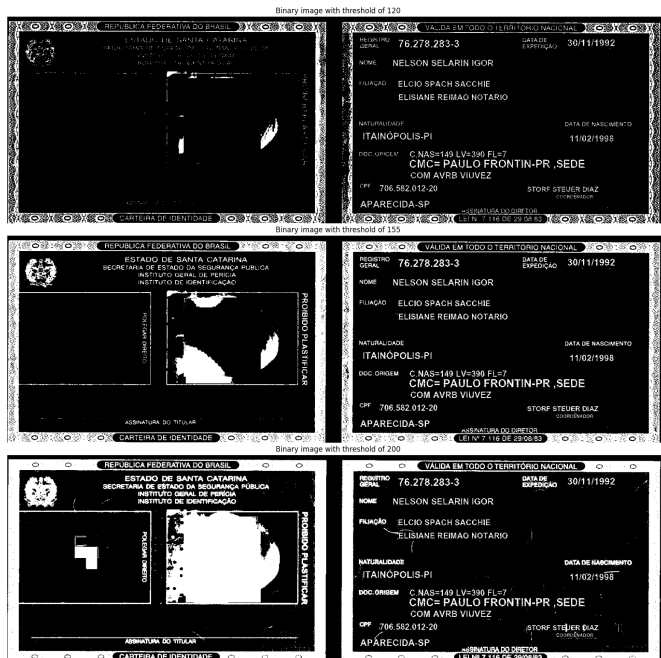


Fig. 3

BINARY IMAGES WITH THRESHOLD VALUES OF 120, 155 AND 200

We can try applying Roberts and Canny filters for edge detection with different values for the threshold1 and threshold2 parameters for the Canny, but the results tend to make the image less smooth for text detection and often introduce additional noise on Canny while the Roberts doesn't add the noise making it a lot better than Canny for this use.

After applying the threshold filter to convert the image to binary, we can extract a specific region and use a dilation morphological filter to enhance the size of the target objects, in this case the letters. This helps improve the text recognition process. Without dilation, the letters often contain small black spots inside. After the first dilation, these spots are noticeably reduced and, with a second dilation, the letter edges become more defined, making them easier to recognize.

With these processes, we obtain a much clearer and smoother image, which facilitates the segmentation process of the text in each field of the document.

The next process is the extraction of part of the document that contains the important information that we want to extract like RG(Registro Geral), CPF(Cadastro de Pessoa Física), Name, Parents, Birth date, and Expedition date. For this process, we can extract the contours and find the one that belongs to the inner section of the document. Cropping the inner part of the document makes it easier for the EasyOCR library to extract the fields of the document without merging



Fig. 4

CANNY FILTERS WITH DIFFERENT THRESHOLD1 AND THRESHOLD2 VALUES.

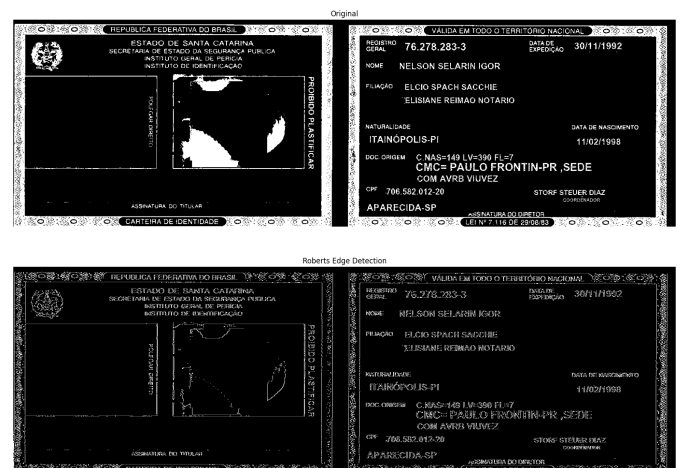


Fig. 5

ORIGINAL IMAGE AND IMAGE AFTER APPLYING ROBERTS FILTER

the boxes with unwanted text fields like "VALIDA EM TODO TERRITÓRIO NACIONAL" and "LEI N°7116 DE 29/08/83".

After the image has been cropped, we pass it through the EasyOCR reader, which will do the segmentation and extraction of the bounding boxes of the fields and then the identification of the words in those fields generating a list of phrases that will be used for identification of fields and extraction of the values of those fields present in the document.

The next process is the extraction of the values of those fields and the filling of Python dictionaries with the information from each field. This process is made with the usage of



Fig. 6

IMAGE SEGMENTS WITHOUT DILATION, WITH 1 DILATION AND WITH 2 DILATIONS



Fig. 7

ORIGINAL IMAGE AND IMAGE AFTER EXTRACTION OF INTEREST ZONE

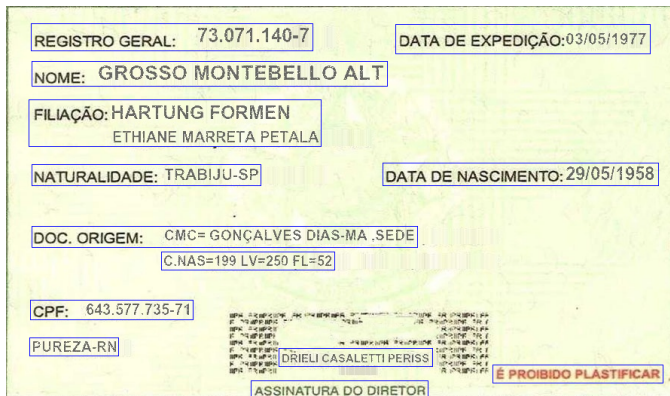


Fig. 8

IMAGE AFTER CROPPING OF INNER REGION AND DETECTION OF THE FIELDS

regular expressions for the search of words that denote each field and their respective values which in case of the dates, RG and CPF which have specific formats making the search process easier.

The last process is to compare the results that the system extracted with the actual values for each field in a group of images. Then look into the accuracy that we were able to get, where we got the most errors in detection and present solutions for improvement.

III. PROPOSED METHOD

To improve the quality of documents and ensure their legibility, this work proposes the use of computer vision techniques focused on image preprocessing, segmentation and

detection of words. Among the techniques used are smoothing filters, edge detection algorithms, and morphological operations. These techniques allow for noise reduction, enhancement of relevant regions, and correction of imperfections that hinder readability. The goal is to improve the clarity and extract the zone of interest of the image content, facilitating reading for the use of automatic recognition systems.

To extract the document information it is important to first segment the fields and then extract the words, that makes it possible to search and assign the values to their respective fields and compare the results with the true values in the documents giving us the possibility to check how accurate the model is and make adjustments to better fit the model and the images that we input it with based on the fields we get most errors of.

IV. EXPECTED RESULTS

It is expected that the application of image processing techniques proposed in this work will result in a significant improvement in the legibility of scanned or photographed documents. With the use of filters, an efficient reduction of visual noise is expected. Edge detection should allow better highlighting of regions of interest, especially textual characters. Morphological operations should contribute to image refinement, correcting flaws, and removing small imperfections. As a final result, the processed images should present greater contrast between text and background, more defined lines, and absence of unwanted elements, enabling clearer reading by users and recognition systems.

It is expected on this paper to apply preprocessing techniques to try to extract the zone of interest for recognition of field values in the documents, and then try to extract the information using EasyOCR python library with the best possible accuracy. In this paper we are aiming for over 75% of accuracy in the detections of field values.

V. RESULTS

First it is necessary to explain how we calculate the errors from the recognitions. The errors are calculated based on the difference between what was assigned as the value of the field by the system in comparison with the real values. This comparisons are made character by character between the two values and for each character wrong another error is summed to the count of errors.

For the results after the whole process of the system, we are able to extract some metrics using a base of 15 images we were able to reach a mean accuracy of 80%. There was one image that was not included on the list of accuracy because the accuracy value that it got was negative because the segmentation got 3 fields in the same box so the algorithm for extraction of values counted the fields and its errors 3 times.

Looking into the graph we can see that 11 out of 14 images got the accuracy above the 75% goal that the system was aiming towards and 3 of them under that goal with 2 of those being a lot below with 31% and 21%, while the other one got closer with 61%.

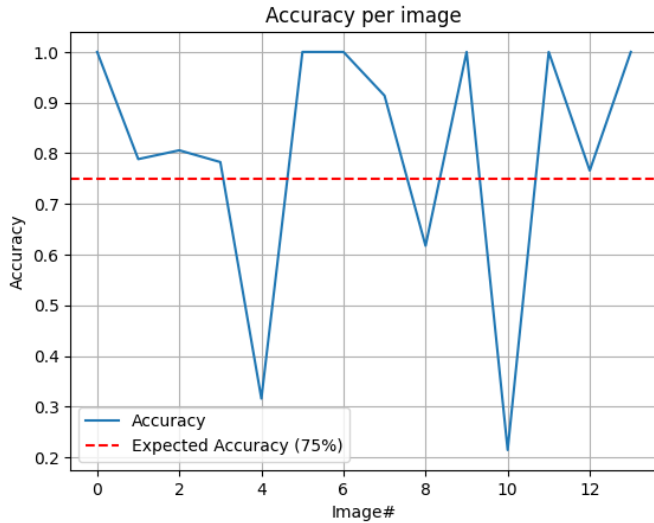


Fig. 9
ACCURACY PER IMAGE

We can also see which fields got more errors as seen in the Figure 10. That graph shows that most part of the errors happens on the "NOME" and "FILIAÇÃO" fields mostly because they are the ones with most letters and looking through the detections we can also observe that in some of the documents we are unable to detect both lines of parents name as the same block so a lot of the letters detected as error are because a part of that field is detected as a different block as we can see on Figures 11 and 12

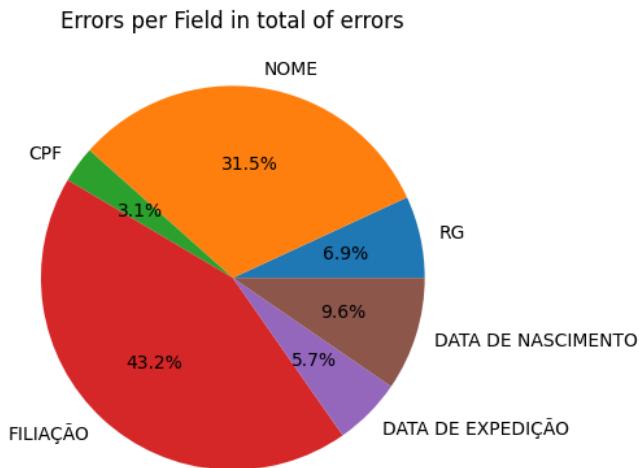


Fig. 10
ERRORS PER FIELD IN TOTAL OF ERRORS

VI. CONCLUSIONS

This project made it possible to apply, in practice, fundamental techniques studied in the Computer Vision course

for processing document images, such as the use of filters, edge detection, and morphological operations, reinforcing the importance of image pre-processing in applications like digitization. It also provided a clear understanding of how simple methods can yield significant results in solving real-world problems. Also the knowledge of OCR showing us how to work with computer vision to extract information from documents generating metrics for analysis of the system.

We have some areas of improvement for the system such as refinement on parameters for EasyOCR reader so we are able to get a better segmentation especially on "Filiação" field, since the names of father and mother are in different lines, which makes the detector miss the name that is underneath, missing information on that field.

Another improvement is in the contours detection for segmentation of the interest zone of the document like in Figure 7. Sometimes the detection of contours used are not able to detect the inner part of the document not making the segmentation possible and sometimes making the segmentation of the "REGISTRO GERAL" and "DATA DE EXPEDIÇÃO" fields harder because of the text that is present at the upper part of the document as shown in Figure 11.

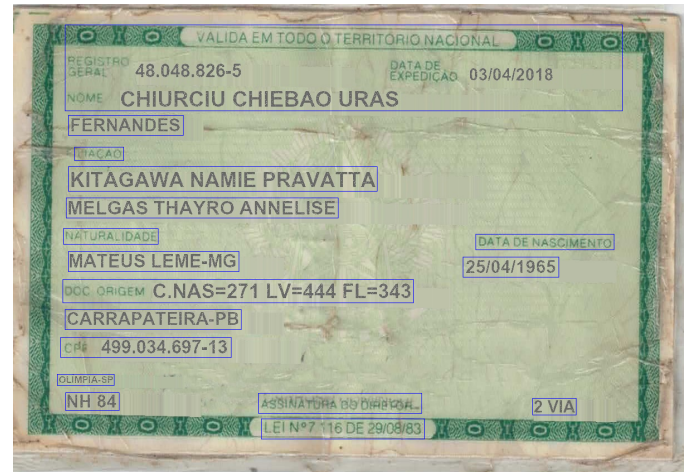


Fig. 11
ERROR ON SEGMENTATION OF FIELDS BECAUSE OF WRONG CROPPING ON THE IMAGE.

The last improvement observed is in the algorithm for extraction of the values from the list generated after the word detection by the reader, for example in the case of multiples fields getting segmented together as one block, the algorithm should be able to extract the values from this block even with segmentation going wrong with the improvement of the algorithm used to search for key words and the values in the text.

REFERENCES

- [1] J. AI. (2024) Easyocr documentation. Accessed: 2025-06-12. [Online]. Available: <https://www.jaied.ai/easyocr/documentation>
- [2] Matplotlib Development Team. (2025) Matplotlib: Visualization with python. Accessed: 2025-06-12. [Online]. Available: <https://matplotlib.org/stable/contents.html>

REGISTRO GERAL	24.102.560-6	DATA DE EXPEDIÇÃO	07/07/2006
NOME	ESPANHOL NAZARRE CONCHETI		
FILIAÇÃO	ELIESER MUKAI KULLER		
	COMARIN TUFANI MAURICIO		
NATURALIDADE	ARAÇAS-BA	DATA DE NASCIMENTO	07/09/1961
DOC. ORIGEM	C.NAS=433 LV=115 FL=97		
PILÕES-PB			
CPF	222.828.412-20		
NH 84	MARENGO ABDUCHI MANHOLER ASSINATURA DO DIRETOR	3 VIA	

Fig. 12

MISSING SECOND LINE ON "FILIAÇÃO" FIELD

- [3] R. P. Fernandes, "Pre processamento e filtros visao computacional," 2025, lecture slides, IBM3131, IBMEC-RJ.
- [4] —, "Pre processamento e filtros 2 deteccao de bordas visao computacional," 2025, lecture slides, IBM3131, IBMEC-RJ.